EXPERIMENT 8: TCP SOCKET COMMUNICATION (REVERSE-SHELL CONCEPT DEMONSTRATION)

AIM:

To study TCP client-server communication using sockets in a controlled localhost environment.

PROCEDURE:

1. Save the server and client programs below into separate files.

2. Run the server first, then run the client on the same computer.

3. Exchange text messages to observe the TCP connection and data transfer.

4. This classroom demonstration intentionally does not execute operating-system commands or provide remote shell access.

5. Check that the server and client use the same IP address and port number.
6. Observe how the server accepts a TCP connection and receives data from the client.
7. Send more than one message to confirm that the connection can carry repeated data exchanges.
8. Enter the exit keyword and verify that both programs close the connection cleanly.

CODE:

```
# tcp_server.py
import socket

HOST = "127.0.0.1"
PORT = 5001

with socket.socket(socket.AF_INET, socket.SOCK_STREAM)
as server:
    server.bind((HOST, PORT))
    server.listen(1)
    print("TCP server waiting on", HOST, PORT)
    conn, addr = server.accept()
```

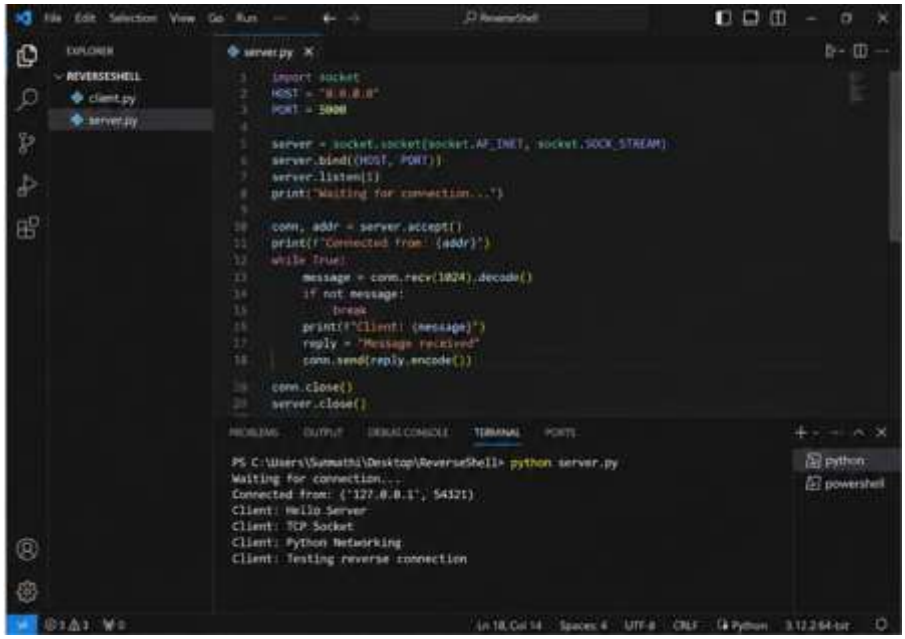
```
    with conn:
        print("Connected by", addr)
        while True:
            data = conn.recv(1024)
            if not data:
                break
            message = data.decode(errors="replace")
            print("Client:", message)
            conn.sendall(("Received: " +
message).encode())

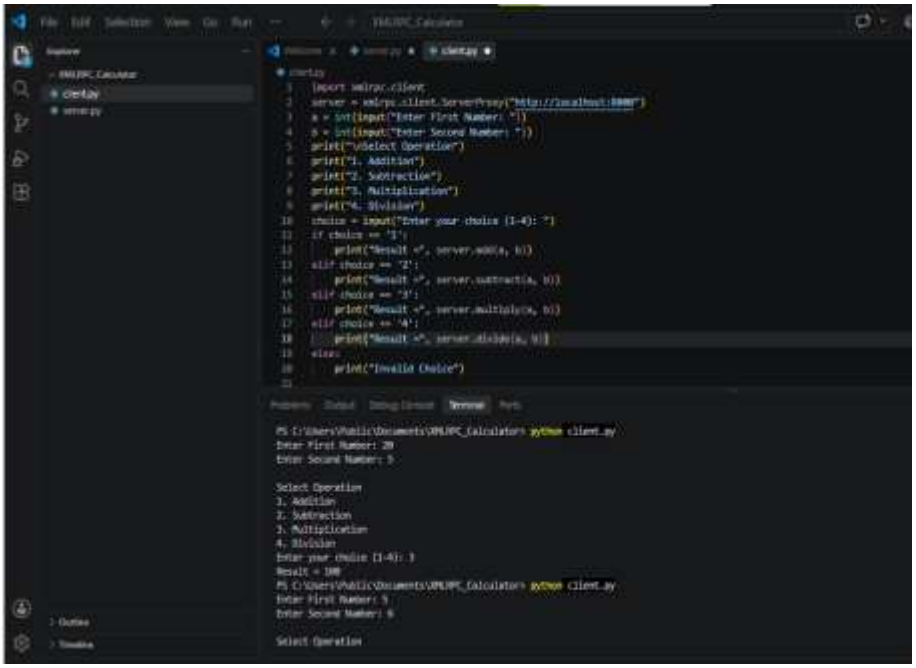
# tcp_client.py (save separately)
import socket

with socket.create_connection(("127.0.0.1", 5001)) as
client:
    while True:
        message = input("Message (quit to exit): ")
        if message.lower() == "quit":
            break
        client.sendall(message.encode())
        print(client.recv(1024).decode())
```

OUTPUT:

:





RESULT:

Thus, TCP socket communication between a client and server was demonstrated in a controlled local environment.